Electrical BUS Literature Review

Overview of the Electrical BUS Subsystem

The Electrical BUS subsystem's goal is to provide the CubeSat with a centralized point of control, which at a high level will involve observing indicators and driving actuators. More specifically, Electrical BUS will interact with the PAYload, RF, and EPS subsystems to manage the state of affairs on the CubeSat. Operations of the electrical subsystem may include directing the PAYload subsystem to send data to the RF subsystem for downlinking, and acquiring uplink data from RF to perform a reboot.

Literature Review Methodology

As it pertains to the design, the main components that make up the electrical subsystem essentially boil down to three categories: control unit (that is, device(s) responsible for carrying out generic operations of the electrical subsystem, for example an MCU), protocol (that is, an interface for connecting two or more devices and allowing them to share data in one way or another, for example I2C), and peripherals (that is, device(s) responsible for a particular function within the Electrical BUS subsystem, for example a watchdog). This classification is not a recognized one, but has been found to be useful when conducting this literature review since the overwhelming majority of the designs can be deconstructed into components each of which would fit into one of these categories [subjective].

In terms of sources, two major categories are surveyed:

- Commercial OBCs discussed in NASA's Small-Sat State of the Art reports from 2022 to 2026 ([1, pp. 215–226], [2, pp. 225–237], [3, pp. 227–239], [4, pp. 241–253]), with trends also observed.
- OBCs built by academic design teams around Canada and beyond (considered designs listed in Appendix A)

For practical interface-throughput characterization, the survey scope is expanded beyond CubeSat-specific implementations where directly comparable measured data are unavailable. Commercial spacecraft hardware, space-industry implementations, semiconductor-vendor benchmarks, and other technically documented tests are considered when they provide both a configured or nominal interface rate and an achieved useful data rate. CubeSat and spacecraft-specific measurements are preferred where available. These broader references are used to characterize throughput efficiency only and do not imply that the referenced hardware is directly suitable for the GOOSE design.

Document Revisions

Version	Date	Author	Description
0.1	July 2026	Boris	Initial
0.2	July 2026	Jennifer	Expanded protocol, interconnect, and COTS trade studies.
0.3	July 2026	Boris	Protection considerations
0.4	July 2026	Sayali	Memory and peripthterals
0.5	July 2026	Matthew	PAY connectors
0.6	July 2026	Asad	SRS-4 integration
0.7	August 2026	Boris	PC/104 pinout survey
0.8	August 2026	Jennifer	PAY integration, protocols, and throughput

Design Space

This section outlines the resulting design space for the Electrical BUS Subsystem based on the corresponding literature review. We establish a space of common control units, protocols and peripthérals for our internal design process.

Control Units

The control-unit design space ranges from low-power microcontrollers to heterogeneous processor–FPGA systems. NASA's State of the Art reports show that most conventional CubeSat OBCs use microcontrollers or embedded microprocessors because command execution, telemetry collection, timekeeping, fault management, and subsystem coordination do not require high computational performance [1, 2, 3, 4]. Higher-performance FPGAs, application processors, and system-on-chip devices are generally introduced when the OBC must also support image processing, autonomy, high-rate payload data, or machine-learning workloads.

Table 1: Candidate OBC control-unit architectures

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Single MCU or processor	One device performs command execution, telemetry, timekeeping, fault management, and subsystem control. Lowest complexity and generally lowest power.	EnduroSat OBC, ARM Cortex-M7 [2, p. 229]; AAC Clyde Space Kryten-M3, Cortex-M3 [2, p. 228].	STM32L4 PC/104 OBC; Athena TMS570; ORCASat TMS570 [5, 6, 7].
Multicore SoC	Multiple cores share one device and memory system. Provides additional performance or lockstep capability, but does not remove the SoC as a single point of failure.	Argotec FERMI, dual-core LEON3FT [2, p. 229]; SPiN MA61C, dual-core GR712RC [3, p. 236]; Xiphos Q7S, dual-core Cortex-A9 [3, p. 238].	Athena and ORCASat use lockstep-capable ARM safety microcontrollers [8, 9].
Dual redundant processors	Two processor devices provide cold, warm, or active redundancy. One processor may take control following a fault in the primary device.	Andes OBC, two Cortex-M7 processors with full cold redundancy [4, p. 230]; Bradford Binary OBC, dual Cortex-R5 with internal cold redundancy [2, p. 229]; Bradford Stellar OBC, dual PolarFire SoCs with internal cold redundancy [4, p. 230].	No directly equivalent two-device redundant architecture was identified in the academic survey.
Controller plus FPGA or DPU	A low-power controller performs critical spacecraft control while an FPGA or higher-performance processor handles payload data or acceleration. The secondary unit may be powered down when unused.	KP Labs Antelope, RM57 OBC plus Zynq UltraScale+ DPU [2, p. 230]; NOVI Gen 2, Vorago MCU plus Versal AI Edge SoC [4, p. 232].	NEUDOSE, primary AVR32 plus secondary Xilinx Zynq-7000 [10].
FPGA-centric	Control or processing is implemented using programmable logic and, optionally, a soft processor. Suitable for deterministic parallel interfaces and high-rate data paths, but more difficult to develop as a general-purpose OBC.	LASP Generic SmallSat FPGA Board, Kintex-7 [2, p. 230]; KP Labs Lion, Kintex UltraScale [2, p. 230].	No FPGA-only primary OBC was identified in the academic survey.

Table 1: Candidate OBC control-unit architectures (Continued)

Architecture	Characteristics	NASA SOA Examples	Academic Examples
Distributed controllers	Subsystem-local processors control the EPS, ADCS, communications, or payload and exchange data with a coordinating OBC. This improves fault containment but increases network and software complexity.	NASA identifies federated, distributed, heterogeneous, and mixed-criticality avionics as viable SmallSat architectures [2, p. 225] [4, p. 221–222].	Individual subsystem controllers used alongside the primary academic OBC.

All architectures will be considered during preliminary design; however, the current baseline is a centralized microcontroller-based OBC with independent watchdog and reset circuitry. This approach is the most common among the academic systems surveyed and provides the lowest-risk balance of power, determinism, software complexity, and flight heritage. An FPGA or secondary processor may be added later if payload-processing requirements cannot be met by the primary MCU.

Protocols and Interconnect

This section will summarize the survey or protocols across the NASA's SOAs as well as Canadian design teams' CubeSats in Table 2, as well as provide an overview of interconnect architectures in Table 3. Communication interfaces differ not only in throughput, but also in topology, wiring complexity, fault containment, and implementation requirements. Consequently, protocol selection should consider the electrical implementation and system architecture in addition to nominal data rate.

Table 2: Protocol and electrical-interface design space

Protocol	Representative Rate	Survey Examples	Primary Considerations
I^2C	100 kbit/s–3.4 Mbit/s	LORIS OBC [11]; identified as a common synchronous interface in all four NASA surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Low pin count and broad peripheral support; limited distance and bandwidth; shared-bus faults may block all devices. Bus capacitance and pull-up sizing constrain the usable rate as devices are added. If SDA remains stuck low, the I^2C specification recommends nine SCL pulses, followed by hardware reset or power cycling if the bus is not released [12].
SPI	Application-dependent; commonly 1–50+ Mbit/s	LORIS OBC [11]; NASA interface surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Deterministic and comparatively fast; requires separate chip-select signals and provides no standardized addressing or error handling. Clock mode, chip-select timing, and SCLK routing or fan-out must be verified for each peripheral, particularly at higher rates [13].

Table 2: Protocol and electrical-interface design space (Continued)

Protocol	Representative Rate	Survey Examples	Primary Considerations
UART	Device-dependent; commonly kbit/s to several Mbit/s	LORIS uses UART [11]; asynchronous serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Simple asynchronous point-to-point communication with low implementation complexity. UART provides serial framing and basic receive-error detection, while subsystem addressing, arbitration, acknowledgements, and recovery policies generally require application-level or higher-level software [14].
RS-422	Approximately 90 kbit/s at 1,200 m to 10 Mbit/s at 4.5 m	Differential serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Differential simplex multidrop communication with one driver and up to ten receivers. It provides greater noise immunity and usable cable length than logic-level UART, but requires appropriate cabling and termination [15].
RS-485	Approximately 100 kbit/s at 1,200 m to 10 Mbit/s at 12 m	Differential serial interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	Differential multipoint communication allowing multiple transceivers to share a bus. Termination is required for reliable higher-rate or longer links, and half-duplex implementations require driver-direction control [15, 16].
CAN / CAN FD	1 Mbit/s Classical CAN; commonly up to 8 Mbit/s CAN-FD data phase	STM32L4 PC/104 OBC [5]; CAN is listed as a common spacecraft fieldbus [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 224–225].	Differential multidrop operation, arbitration, acknowledgements, and error confinement make it suitable for EPS, ADCS, RF, and housekeeping traffic. Payload throughput is limited. The bus should be terminated at its two physical ends, with node stubs kept short to reduce reflections [17, 18].
USB 2.0	12 Mbit/s full-speed; 480 Mbit/s high-speed	LORIS OBC [11].	High COTS compatibility and throughput, but host/device roles, enumeration, connector retention, software complexity, VBUS sensing, and the board's USB power arrangement must be defined [19, 20].
Ethernet	100 Mbit/s–1 Gbit/s commonly	Listed as a network interface in each NASA survey [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224].	High throughput and broad software support; requires switching or routing and may require additional measures for deterministic real-time operation.

Table 2: Protocol and electrical-interface design space (Continued)

Protocol	Representative Rate	Survey Examples	Primary Considerations
SpaceWire / SpaceFibre	SpaceWire: 2–400 Mbit/s; SpaceFibre: up to 5 Gbit/s per lane and over 20 Gbit/s using multiple lanes [21, 22]	SpaceWire is listed in the 2022–2024 surveys [1, p. 219] [2, p. 235] [3, p. 240]; SpaceWire and SpaceFibre are included together in 2026 [4, p. 225].	Spacecraft-specific point-to-point networking with packet routing and link-level fault management. SpaceFibre additionally supports virtual channels, quality of service, and improved fault detection, isolation, and recovery. Both require more specialized implementation and test resources than conventional MCU buses [21, 22].
PPS (Pulse Per Second)	1 pulse/s	GNSS receivers commonly provide a configurable timing-pulse output; the ZED-F9P is one representative implementation [23].	Provides a precise synchronization edge but does not independently encode the absolute time. It is normally paired with UART or another data interface; electrical level, jitter, routing, and holdover behaviour must be considered.
SerDes / MGT	Gigabit-per-second class; XAUI provides 10 Gbit/s aggregate	Identified by NASA as high-rate mission-data interfaces [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Suitable for FPGA, payload, mass-memory, and radio data paths. Requires controlled-impedance routing, suitable connectors, reference clocks, and signal-integrity analysis.
GPIO / PWM / ADC / DAC	Application-dependent	LORIS provides GPIO, PWM, and ADC [11]; discrete and analog interfaces are listed by NASA [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Appropriate for enables, inhibits, deployment signals, and analog telemetry; creates dedicated wiring and provides little inherent fault detection.
JTAG / ISP / ICSP	Not a mission-data interface	Debug and programming interfaces are listed in the NASA surveys [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Required for programming and board-level debugging; access should be isolated or protected during flight.

Higher-Level Communication Protocols

The interfaces listed in Table 2 describe the electrical and link-level means by which data is carried between subsystems. Higher-level communication protocols can additionally define how messages are addressed, routed, organized, and delivered across those interfaces.

CubeSat Space Protocol (CSP) is a lightweight packet-oriented protocol stack intended for distributed embedded systems. It provides subsystem node addressing, port-based services, packet routing, and datagram transport. CSP supports interfaces including CAN, I^2C, serial links using KISS over RS-232 or RS-422, and Ethernet [24, 25]. This allows a common communication structure to be used across multiple subsystem links. The project would still need to define its node addresses, routes, application services, and whether reliable delivery is required.

Remote Memory Access Protocol (RMAP) is a standardized protocol that operates over SpaceWire. It supports reading from and writing to memory in a remote SpaceWire node and may also be used for network configuration, node control, and data transfer [26]. RMAP is therefore relevant when SpaceWire is selected and remote-memory-style access is useful, rather than being a general replacement for communication over CAN, UART, or other interfaces.

A project-specific packet format remains another possible approach and may reduce implementation complexity for a small network, but would provide less standardization and interoperability than an established protocol. Selection of a higher-level protocol should therefore consider subsystem compatibility, software complexity, reliability requirements, packet overhead, and the capabilities of the underlying interface.

Table 3: Candidate interconnect architectures

Architecture	Connection Pattern	Advantages	Limitations
Unified stack	OBC, EPS, RF, ADCS, and payload share a PC/104-style stack-through connector carrying power and common buses. The STM32L4 academic OBC uses this format [5]. NASA discusses PC/104 and stackable mezzanines in [1, p. 210], [2, pp. 226–227], [3, pp. 229–230], and [4, p. 222].	Compact, modular, rugged, and requires little harnessing. The self-stacking arrangement eliminates the need for a separate backplane or card cage [27].	Shared pinout constrains all boards; board order, connector bandwidth, and available signal lanes may limit expansion. Shared power or bus paths can also reduce fault isolation between boards.
Backplane	Subsystem cards plug into a shared CompactPCI or SpaceVPX backplane [1, p. 210] [2, p. 226] [3, p. 229] [4, p. 222].	Rigid mechanical support, modular replacement, standardized slot and module interfaces, and high-speed routing. SpaceVPX additionally targets interoperable and fault-tolerant systems [28].	Higher mass, volume, connector cost, and chassis complexity, with additional mechanical, electrical, and interoperability verification.
Centralized point-to-point	Each subsystem has a dedicated connection to the OBC; all commands and data pass through the OBC. NASA identifies direct point-to-point and centralized architectures in [1, p. 207] and [3, p. 226].	Simple command authority, dedicated links, predictable data paths, and strong link-level isolation.	Connector and harness count increase with the number of subsystems. The OBC can become a bandwidth bottleneck and single point of failure, particularly when high-rate payload traffic passes through it [4, pp. 221–223].
Shared peer network	OBC, EPS, RF, ADCS, and payload controllers communicate as peers over CAN, Ethernet, or SpaceWire.	Reduces dedicated wiring, allows direct subsystem communication, and supports distributed control. Subsystem-local processing may reduce dependence on the primary OBC [4, pp. 221–222].	Requires addressing, arbitration or routing, synchronization, and network-level fault management. Shared-bus behaviour, node failures, and any redundancy-management strategy must be defined and verified [18, 21].
Payload pass-through	The OBC controls the payload, while high-rate data travels directly between the payload, mass memory, FPGA, and RF subsystem.	Avoids loading the housekeeping bus or copying all payload data through the primary MCU, while allowing a dedicated path to support higher throughput [4, pp. 222–223].	Requires separate control and data paths, flow control, buffering, and an OBC mechanism to isolate or reset the data path. The OBC must retain sufficient status and fault visibility to manage the path safely.

Table 3: Candidate interconnect architectures (Continued)

Architecture	Connection Pattern	Advantages	Limitations
Hybrid dual-plane	A low-speed CAN or stack bus carries commands and housekeeping data, while dedicated Ethernet, SpaceWire, or SerDes links carry payload data. NASA describes this mixed topology in [4, pp. 222–223].	Combines a robust housekeeping network with scalable payload throughput. Separating control and high-rate traffic can reduce contention and improve fault containment between traffic classes.	Requires two interface classes, additional integration and verification, and clear ownership of commands, telemetry, timing, and high-rate data. Fault detection and recovery must be coordinated across both planes.

Redundancy is not necessarily a separate base topology and may instead be applied selectively to critical links, buses, or nodes within any of the architectures above. It can improve availability, but adds transceivers, wiring, connector pins, power consumption, software logic, and verification effort [4, 18].

For the mission, the interconnect as well as the protocol bouquet selection will want to target higher throughput interfaces for RF and PAY subsystems, and thus will generally be dictated by them. In the spirit of a generic design, selecting a higher-speed interface is favoured, yet this might collide with the interconnect architecture selection since certain interfaces require specific connections that may not, for example, be routed through PC/104 due to signal integrity requirements (e.g. SpaceFibre). The stack-up requirements and/or constraints might also be driven by Mechanical subsystem. Consequently, interface selection should balance throughput with connector capability, harness length, routing, and signal-integrity constraints rather than relying on nominal data rate alone [21, 22].

Peripherals

This section reviews the common peripherals that are found on OBCs surveyed by NASA's SOAs, as well as on OBCs designed by Canadian academic design teams. Details about volatile and non-volatile memory can be found in Tables 4 and 5, respectively. Other types of peripherals are summarized in Table 6. General considerations related to peripherals revolve around external vs internal peripherals since modern day SoCs contain many of the necessary peripherals on the same chip.

Volatile Working Memory

Volatile working memory, which loses functionality when the electrical power is disrupted, acts as high-speed temporary storage. For a CubeSat, it enables real-time software tasks. The primary consideration during implementation is between internal and external memory: internal memory (built within the main processor chip) reduces component count and PCB complexity; external memory provides greater capacity but introduces additional power and routing requirements. Other features such as Error-Correcting Code (ECC) can be chosen as needed; ECC RAM, which automatically detects and corrects single-bit errors, was used by ORCASat [9].

Table 4: Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
SRAM (Static RAM)	Typically internal cache or memory within an SoC, used for low-latency code execution. Surveyed by NASA [1, p. 218] [2, p. 234] [3, p. 239] [4, p. 223].	10-25 ns access time [4]	4-32 Mb [4]
DRAM (Dynamic RAM)	High-density external volatile memory that uses single-transistor, single-capacitor cells. Surveyed across NASA missions [1, p. 218] [2, p. 234] [3, p. 239] [4, p. 223].	25 ns access time [4]	N/A

Table 4: Volatile Memory Types, Metrics, Considerations, and Examples (Continued)

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
SDRAM (Synchronous DRAM)	High-density external memory synchronized with the system clock; utilized externally by RADSAT-SK for general runtime tasks [29].	N/A	N/A
DDR3 (Double Data Rate 3)	High-speed third-generation SDRAM; Dalhousie University's LORIS uses 256 MB DDR3 [11].	13-15 ns [30, pp. 67-70]	4 GB [30, p.1]

Non-Volatile Program and Mass Memory

Table 5: Non-Volatile Memory Types, Metrics, Considerations, and Examples

Memory Type	Description, Considerations, & Examples	Estimated Speed	Estimated Capacity
MRAM (Magnetoresistive)	High-endurance, radiation-tolerant non-volatile memory that relies on magnetic states to store data;typically used as internal or external storage for persistent system state. Used by the C3S OBC [2, p. 229] and in redundant configurations on the Andes OBC [3, p. 232] [4, p. 230].	300 ns access time [4, p. 224]	up to 8 Gb [31]
FRAM (Ferroelectric RAM)	Low-power, fast-write non-volatile memory (internal or external) that stores memory due to its ferroelectric properties; electric dipoles stay in place even without power [2, p. 229].	300 ns access time [4, p. 224]	up to 16 Mb [31]
NAND Flash	External storage where memory cells are connected in series; deployed in redundant configurations on the Andes OBC [3, p. 232] [4, p. 230].	25–50 μs read access time [32]	128 Mb–2Tb [33]
NOR Flash	External storage where memory cells are connected in parallel; used by Athena [6] and ORCASat [7].	10–100 ns read access time [32]	128 Mb–2 Gb [33]

Non-volatile mass memory is persistent computer storage that retains stored data without requiring a continuous power supply, unlike volatile memory. It can store system boot code, flight software images, and long-term scientific data. Unlike the considerations for volatile memory, selection for non-volatile memory depends on many factors, including capacity, write endurance, retention, access speed, boot reliability, and susceptibility to radiation-induced corruption. Redundant boot images and error detection should be considered. After choosing a memory type, the packaging can be considered. For example, an eMMC (embedded MultiMediaCard) is a high-density external mass storage featuring an integrated flash controller; used by the C3S OBC [2, p. 229] and by LORIS (4 GB eMMC) [11].

Table 6: Common OBC peripheral classes

Peripheral Class	Examples	Primary Considerations
Watchdog, reset, and supervisor circuitry	The C3S OBC uses a multi-level watchdog [2, p. 229] [3, p. 232] [4, p. 230]; the Aerospacelab OBC includes a radiation-hardened supervisor MCU and watchdog [3, p. 232] [4, p. 229]. NASA discusses watchdogs and power cycling as fault-mitigation mechanisms [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 220–222].	An external watchdog is preferable for recovery from processor lock-up. Consider independent clocks, reset coverage, timeout configuration, communications watchdogs, and selective power cycling.

Table 6: Common OBC peripheral classes (Continued)

Peripheral Class	Examples	Primary Considerations
Timers and time-keeping	NASA identifies PLLs, capture/compare units, and time-processing units as common OBC peripherals [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 224]. Timers are integrated within the STM32L4, TMS570, and i.MX6DL devices used by the surveyed academic OBCs [5, 8, 9, 34].	Used for scheduling, timestamping, watchdog servicing, and waveform generation. Clock drift, oscillator redundancy, synchronization, and operation during processor resets must be considered.
Analog and discrete I/O	NASA lists GPIO, ADC, and DAC peripherals in each avionics survey [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 224–225]. LORIS exposes GPIO, PWM, and ADC interfaces [11, 34].	Used for analog telemetry, enables, inhibits, deployment signals, and simple sensors. Consider voltage compatibility, input protection, ADC accuracy, safe reset states, and whether signals require isolation or redundancy.
Debug and programming interfaces	NASA lists JTAG, ISP, ICSP, BDM, BITP, and DB9 interfaces [1, p. 219] [2, p. 235] [3, p. 240] [4, p. 225].	Required for programming, boundary scan, manufacturing tests, and debugging. Flight hardware should provide controlled physical access and prevent unintended entry into debug or programming modes.
Optional integrated navigation peripherals	The EnduroSat OBC is available with integrated GNSS [2, p. 229] [4, p. 230].	Integration can reduce board count and harnessing, but may couple the OBC lifecycle to a specific receiver and complicate antenna placement, RF isolation, replacement, and independent fault recovery.

[TODO: survey each class of peripherals in more detail]

Connections between PAY and BUS

Integration with PAY is probably the most significant aspect of the OBC design since supporting the PAY is necessary for the success of the mission in any capacity. The design should be generic enough to fit a wide array of payloads, therefore commecial standalone OBCs are considered during the review since these are essentially what we strive to mimic with our design. The following considerations are of interest:

1. **Data and control communication throughputs** – that is, at what rate the OBC and PAY can exchange data.
2. **Buffers** – that is, how much spare volatile and non-volatile memory the OBC provides to PAY to store its telemetry and mission data, as well as the OBC's own data.
3. **Transfer mechanism** – that is, how PAY will request the OBC to let it access the data for read or write, and how it will be notified of other control-level events.

Practical Interface Throughput

The representative rates in Table 2 describe nominal, configured, or typical interface capabilities. The useful data rate available to PAY at the application level can be lower because transmitted traffic also contains framing, addressing, error-detection information, acknowledgements, inter-frame gaps, and higher-level protocol overhead. Processing, buffering, DMA availability, bus contention, and software implementation may introduce additional limitations.

For comparison, an interface-throughput ratio is defined as

$$\eta_{\text{throughput}} = \frac{R_{\text{useful}}}{R_{\text{nominal}}}, \tag{1}$$

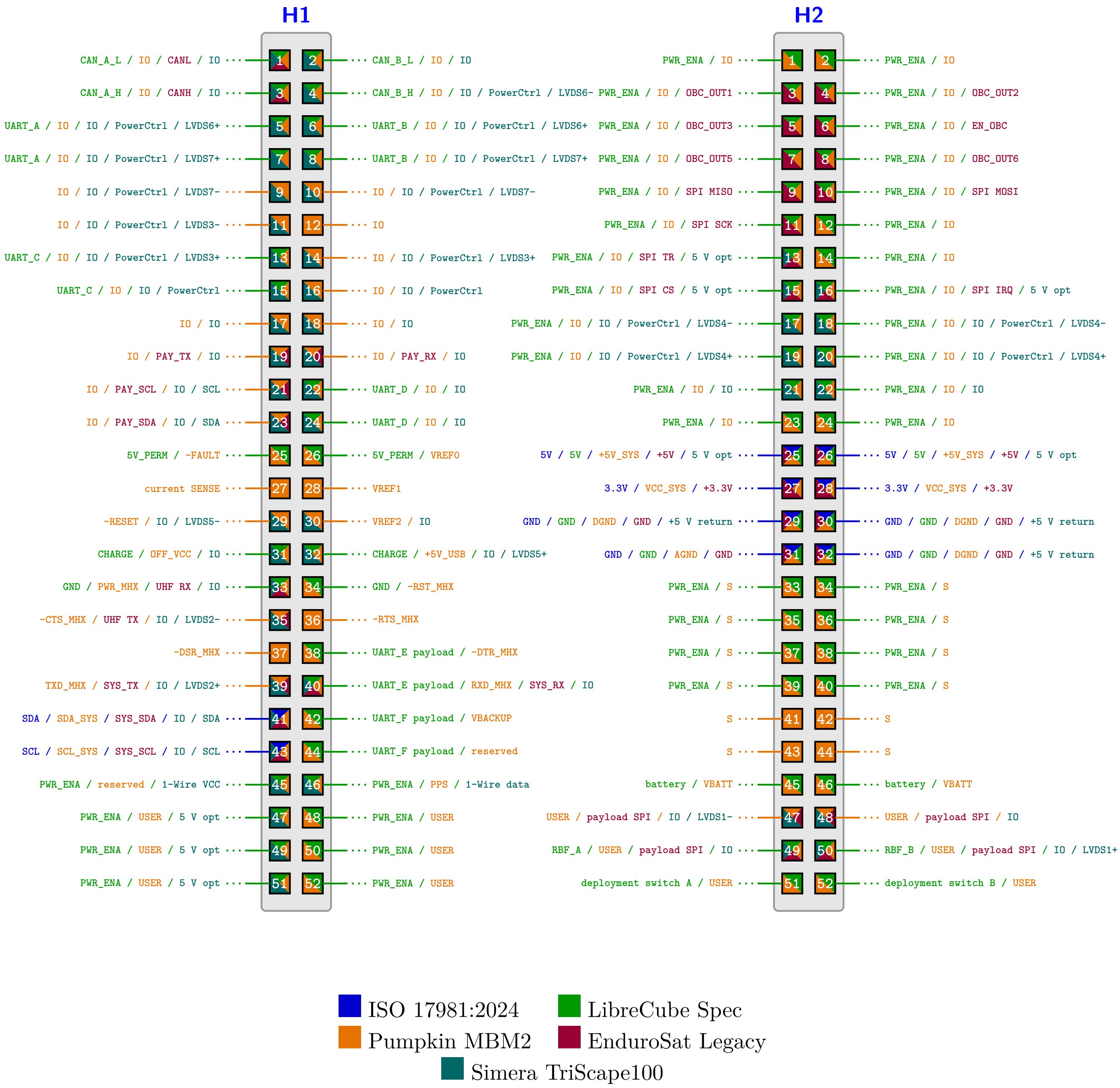

Figure 1: Overlay of various standards and OBC pinouts using PC/104 headers

where R_{useful} is the demonstrated useful or application-level data rate and R_{nominal} is the corresponding configured, signalling, or Layer-1 rate. This ratio is implementation-dependent and should not be treated as a universal protocol constant. Table 7 therefore records the operating conditions associated with each reference value.

Table 7: Reference practical throughput relative to nominal or configured interface rate

Interface / Implementation	**Nominal / Configured Rate**	**Demonstrated Useful Rate**	**Ratio**	**Condition and Source**
CAN (SRS-4, CSP)	1.00 Mbit/s	0.45 Mbit/s	45.0%	Satlab application-layer goodput using CSP over CAN. The same value is reported for 217-byte and 1024-byte application frames [36].
Classical CAN	500 kbit/s	≈281 kbit/s	≈56.2%	OTH Regensburg/CiA measured implementation using an 8-byte user payload and 227.6 μs end-to-end message-processing time [?].
CAN FD	1 Mbit/s arbitration; 4 Mbit/s data phase	≈2.46 Mbit/s	≈61.5%*	OTH Regensburg/CiA measurement using a 64-byte user payload and 208.3 μs end-to-end message-processing time [?].
RS-422 (SRS-4, CSP)	12.5 Mbit/s	9.47–9.74 Mbit/s	75.8–77.9%	Satlab application-layer goodput. The lower value uses 217-byte frames and the upper value uses 1024-byte frames [36].
RS-485	Implementation-dependent	—	—	No directly comparable application-goodput percentage was identified in the reviewed sources. RS-485 specifies the electrical layer, so useful throughput depends on the serial framing, packet protocol, bus loading, and half-duplex turnaround used by the implementation [15].
UART	128 kbaud	86.2 kbaud	≈67.3%	TI TMS320TCI6616 internal-loopback measurement with one start bit, eight data bits, one parity bit, and one stop bit; L2 memory was used as the source and destination [?].
I^2C	400 kHz SCL	351.7–354.5 kbit/s	87.9–88.6%	TI TMS320TCI6616 internal-loopback benchmark using EDMA and 256–1024-byte read/write transfers [?].
SPI	66 Mbit/s	57 Mbit/s	≈86.4%	TI TMS320TCI6616 SPI benchmark using DMA-assisted transfers; the documented implementation includes idle cycles between words [?].
USB 2.0 High-Speed Bulk	480 Mbit/s	43.8 MB/s (350.4 Mbit/s)	≈73.0%	Cypress/Infineon FX2LP measured bulk-transfer test on an Intel ICH9 USB 2.0 host running Windows 7, with data generated internally by the FX2LP [?].
Ethernet (SRS-4, UDP/IP)	100 Mbit/s	53.04–84.30 Mbit/s	53.0–84.3%	Satlab application-layer goodput for 217-byte and 1024-byte frames, respectively. Satlab identifies these values as CPU-limited rather than protocol-overhead-limited [36].

Table 7: Reference practical throughput relative to nominal or configured interface rate (Continued)

Interface / Implementation	Nominal / Configured Rate	Demonstrated Useful Rate	Ratio	Condition and Source
SpaceWire	200 Mbit/s signalling	152–160 Mbit/s	76–80%	STAR-Dundee reference for current radiation-tolerant SpaceWire technology: approximately 160 Mbit/s unidirectional or 152 Mbit/s bidirectional per link [?].
SpaceFibre	6.25 Gbit/s/lane	≈4.6–4.8 Gbit/s	≈73.6–76.8%	STAR-Dundee STAR-Ultra single-lane router reference: approximately 4.8 Gbit/s unidirectional or 4.6 Gbit/s bidirectional per port [?].

** The CAN FD ratio is shown relative to the 4 Mbit/s data-phase rate. The arbitration phase operates at a separate 1 Mbit/s rate, so this ratio is not directly equivalent to the single-rate interface ratios.*

The surveyed results do not support applying one universal utilization factor to all interfaces. In the referenced implementations, useful throughput ranges from approximately 45% of the configured rate for CSP over CAN on the SRS-4 to approximately 80–90% for several sustained point-to-point or long-transfer interfaces. Differences arise from framing, payload size, arbitration or bus sharing, processing architecture, DMA support, and software overhead. These ratios should therefore be treated as reference points for preliminary GOOSE sizing rather than guaranteed performance. The selected GOOSE implementation should be benchmarked directly when PAY throughput becomes a driving requirement.

Throughput Margin

Demonstrated goodput and engineering margin are treated separately. Goodput accounts for protocol and implementation losses, while engineering margin reserves unused capacity for requirement growth, contention, timing variation, and other uncertainties. NASA software-engineering guidance provides historical Marshall Space Flight Center examples of data-bus-throughput margins of 35% at CDR, 25% at TRR, and 20% at launch, while cautioning that the appropriate margin depends on how the resource is defined and on project context [?]. These values are reference guidance and are not adopted as GOOSE requirements at this stage.

A preliminary allocated rate may therefore be represented as

$$R_{\text{allocated}} = R_{\text{useful}}(1 - M), \tag{2}$$

where M is the selected project throughput margin.

Payload Buffering Considerations

At the system level, PAY data may be held temporarily in volatile memory before being processed, transferred to non-volatile mass storage, or forwarded to the RF subsystem. Volatile memory therefore provides short-term working space for incoming data, while non-volatile memory can retain larger data products until they can be processed or downlinked. As shown in Table 14, commercial OBCs commonly provide both forms of memory; however, their total installed memory does not necessarily represent the amount that can be allocated specifically to PAY.

The required buffering depends on the PAY data pattern. Burst-oriented payloads, such as imaging systems, require enough temporary capacity to retain the largest expected data product until it can be transferred. Continuous payloads instead require the sustained processing, storage, or forwarding rate to remain comparable to the incoming data rate, since buffering can only absorb temporary rate differences. PAY integration requirements should therefore identify the expected maximum burst size, sustained data rate, whether PAY can pause its transmission, and whether PAY can retain data locally when the OBC is not ready.

A comprehensive survey of commercial OBC platforms and their payload integration parameters is provided in Appendix B (Table 14). Figures 2, 3, and 4 summarize the data rates and memory capacities observed across commercial platforms.

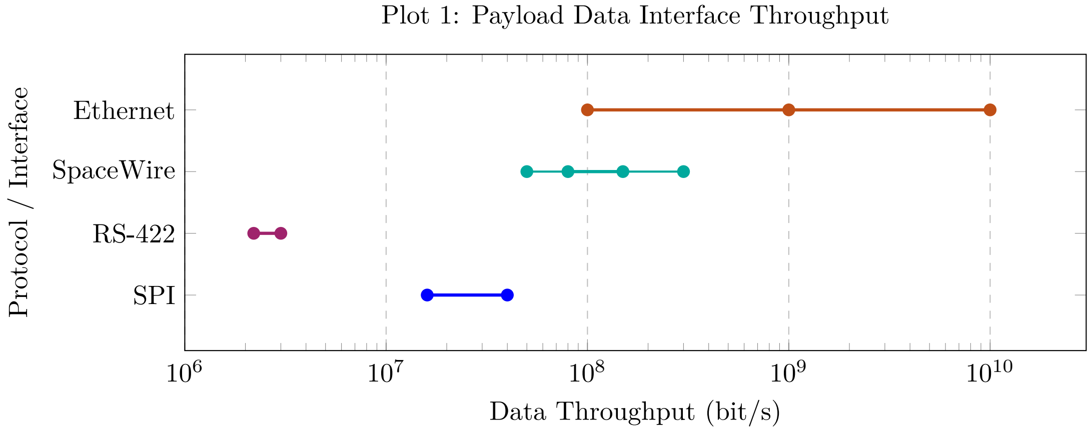

Figure 2: Published payload data interface rates from the commercial OBC survey (Appendix B). Values may be measured, manufacturer-stated, or nominal depending on the source; dots represent specific published figures and lines represent observed ranges. Practical application-level throughput is considered separately in Table 7.

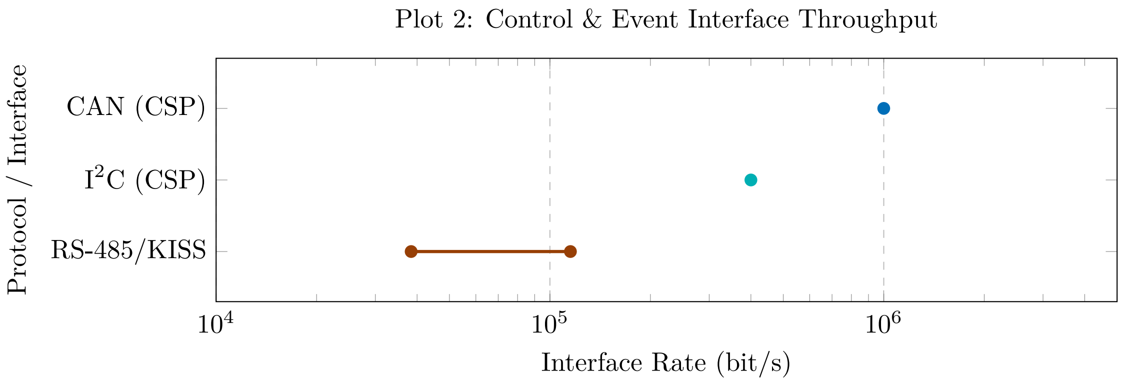

Figure 3: Published control and event interface rates from the commercial OBC survey (Appendix B). These values are configured or published interface rates and should not be interpreted directly as application goodput; see Table 7.

Below in Table 8 is a survey of the OBC integration done by KP Labs.

Table 8: Physical connectors (surveyed from KP Labs Intuition-1 integration [35])

Connector Type	**Supported Protocols**	**Notes**
Nano-D Connectors	LVDS	Used for direct high-speed data transfer (e.g., connected to X/S-band radio).
M80 Connectors	Controller Area Network (CAN) bus	Used to connect DPU supervisor module to platform OBC.
Samtec LSHM	LVDS	Used for high-speed data streams (up to 8 Gbps) between optical instruments and data processing unit (DPU).

[TODO: review more examples]

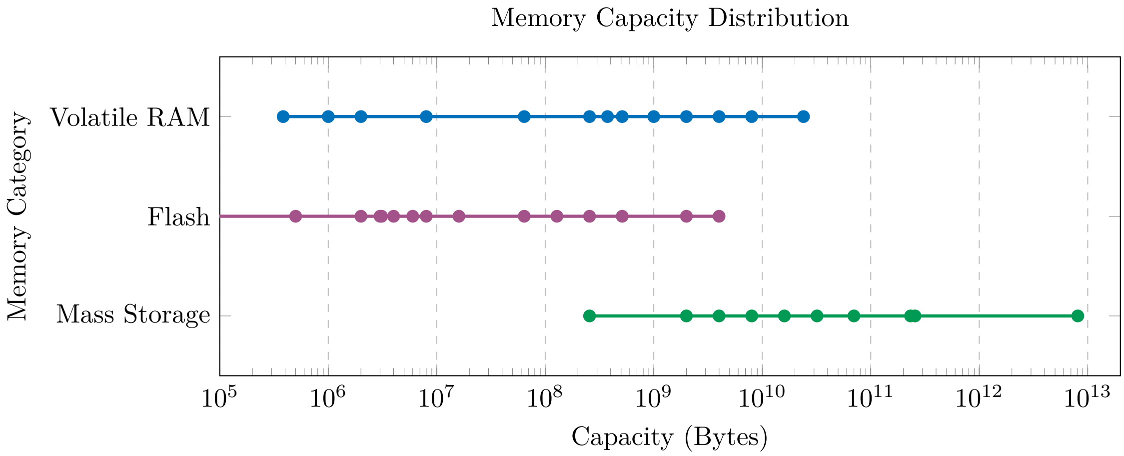

Figure 4: Observed memory capacity ranges and data points across surveyed commercial OBCs: Volatile RAM (384 kB–24 GB), Flash/Boot NVM (32 kB–4 GB), and Non-Volatile Mass Storage (256 MB–8.1 TB).

SRS-4 Integration

The Satlab SRS-4 is a flagship software-defined S-band transceiver designed for CubeSat high-speed communications [36]. The SRS-4 serves as a primary communications radio to support high-throughput payload downlink requirements (such as large hyperspectral image files) alongside housekeeping telemetry and telecommand handling.

The transceiver functions as the satellite's primary voice-and-data translator between the On-Board Computer (OBC) and the RF antenna array. Satlab manufactures the SRS-4 board using the PC/104 form factor ($93.0 \times 87.2 \times 17.5$ mm, mass 253 g), allowing direct mechanical integration into standard CubeSat stackups via PC/104 header connectors and standoffs alongside EPS and OBC modules.

Unlike legacy fixed-function analog radios, the SRS-4 utilizes a Software-Defined Radio (SDR) architecture with on-orbit reconfigurable modulation and full-duplex operation, enabling simultaneous transmission and reception (requiring strict RF isolation between TX and RX paths).

Electrical and Logical Interfacing

The logical and protocol fabric of the SRS-4 natively integrates with standard Command & Data Handling (C&DH) firmware architectures:

- **CubeSat Space Protocol (CSP):** The SRS-4 supports CSP for command, control, and data exchange with the satellite's C&DH system [36]. CSP provides node addressing, port-based services, packet routing, and datagram transport across supported interfaces including CAN, I^2C, serial KISS links, and Ethernet [24, 25]. This allows the radio to operate as a networked subsystem while leaving the underlying electrical link and application services to be defined during system integration.
- **Interfacing Links:**
  - **OBC ↔ SRS-4:** CAN-bus or RS-422 differential bus carrying CSP packets.
  - **EPS ↔ SRS-4:** Wide-input power rail (5.1–28.8 V), granting flexibility to EPS regulation strategies.
  - **SRS-4 ↔ Antenna:** SMA/MCX coaxial pigtail point-to-point RF link.
  - **Antenna ↔ Ground:** Free-space S-band RF link (2200–2290 MHz downlink, 2025–2110 MHz uplink).

Interface Throughput

The SRS-4 datasheet provides a direct product-specific comparison between physical interface rate and application-layer goodput. CSP over CAN provides approximately 0.45 Mbit/s of application goodput from a 1 Mbit/s Layer-1 rate, or approximately 45%. CSP over RS-422 provides 9.47 Mbit/s with 217-byte frames and 9.74 Mbit/s with 1024-byte frames from a 12.5 Mbit/s Layer-1 rate, corresponding to approximately 75.8–77.9% [36].

The same datasheet reports 53.04 Mbit/s and 84.30 Mbit/s of UDP/IP application goodput over a 100 Mbit/s Ethernet interface for 217-byte and 1024-byte frames, respectively, and identifies those Ethernet results as CPU-limited rather than protocol-overhead-limited [36]. These figures illustrate why the nominal interface rate alone is not sufficient for GOOSE data-path sizing; frame size, higher-level protocol, and processing architecture can materially change the rate available to the application.

Subsystem Team Responsibilities

Integrating the SRS-4 into a satellite design spans multiple engineering domains:

- **Electrical Subsystem:** Mechanically slots the SRS-4 board into the PC/104 stack and wires the physical CAN/RS-422 bus and EPS power connections.
- **C&DH Subsystem:** Provides the RTOS and CSP network fabric where the SRS-4 operates as a standard CSP node on the system bus.
- **Firmware Team:** Develops device drivers using Satlab's support library to queue image payload downlinks, adjust TX power, and poll operational telemetry (temperature, current draw, lock status).
- **Power (EPS) Subsystem:** Manages the input power supply rail, leveraging the SRS-4's broad 5.1–28.8 V input range.
- **Mission Operations Team:** Utilizes CCSDS-compliant coding and standard PSK modulations to interface with commercial ground station networks and feed telemetry directly into the mission control framework.

COTS (Commercial Off the Shelf) Design Alternatives

Description: Determine what COTS solutions exist if we cannot produce a certain part of the subsystem in house. Find a COTS option for each in house design.

Will inform trades later on for COTS versus in house risks.

This section outlines alternative off-the-shelf components that would in part or in full fulfil the operating requirements of the subsystem.

The following component-level COTS alternatives preserve a custom Electrical BUS architecture while reducing development effort for selected functions. They are trade-study candidates rather than flight-qualified selections; mission-specific part assurance, derating, and radiation assessment remain required [37].

Table 9: Preliminary component-level COTS alternatives

Function	Example Candidates	Potential OBC Use	Main Trade Considerations
Non-volatile storage	Everspin MR25H40 SPI MRAM for critical state and logs [38], and Infineon S25HL512T serial NOR flash for firmware storage or recovery images [39].	Persistent configuration, deployment state, reset history, fault logs, firmware, and a recovery image.	Capacity, endurance, retention, write protection, boot recovery, power-loss behaviour, and radiation response.
Independent fault recovery	TI TPS3431 watchdog, TPS3703 voltage supervisor, and TPS22919 load switch [40, 41, 42].	Detect processor lock-up, supervise the supply, and provide a hardware reset or power-cycle path.	Independence from the MCU, timeout and threshold selection, reset coverage, load current, default state, and control ownership.

Table 9: Preliminary component-level COTS alternatives (Continued)

Function	Example Candidates	Potential OBC Use	Main Trade Considerations
Timing and clocking	Micro Crystal RV-3028-C7 RTC [43], plus an industrial crystal or TCXO selected using oscillator guidance [44].	Maintain timestamps through resets and provide stable MCU and interface timing.	Backup supply, drift, calibration, startup margin, temperature stability, synchronization, and PCB placement.
Differential bus interface	TI TCAN1042-Q1 CAN/CAN-FD transceiver [45] or Analog Devices ADM3065E RS-485 transceiver [46].	Implement the physical layer between MCU logic and a differential bus.	Common-mode range, delay, standby state, termination, fault protection, isolation needs, and radiation assessment.
Board-health monitoring	TI TMP117 temperature sensor and INA238 current, voltage, and power monitor [47, 48].	Provide local temperature and supply-rail telemetry for health monitoring.	Placement, range, shunt sizing, conversion time, addressing, power, failure response, and radiation behaviour.

System Architecture and Budgets

In this section we concern ourselves with system-level considerations such as power draw, protection mechanisms, and radiation tolerance for the purposes of overall design and integration with other teams.

Power draw

Based on the commercial onboard-computing systems surveyed in NASA's State of the Art reports [1, 2, 3, 4], OBC power consumption can be divided into three broad categories according to platform complexity and computational capability. The reports do not use a consistent average-power metric; reported values are variously identified as idle, static, typical, active-mode, full-load, or peak consumption. Consequently, Table 10 preserves the operating-state terminology used by the manufacturers.

The 2022 edition identifies representative onboard-computing products but does not tabulate their power consumption [1, Table 8-1, pp. 214–217]. Quantitative power values are therefore drawn primarily from the 2023, 2024, and 2026 editions [2, Table 8-1, pp. 228–233] [3, Table 8-1, pp. 230–236] [4, Table 8-4, pp. 229–234].

Table 10: OBC power-consumption trends and reported operating states

Platform Complexity	Reported Power Profile	Design Characteristics	Example Designs
Low (Bare-Metal / Control)	**Minimum listed:** 0.05 W **Idle/steady:** approximately 0.2–0.5 W **Peak:** generally below 1 W	Simple command and data handling, housekeeping, telemetry, and fault management. These systems are normally based on low-power microcontrollers and execute bare-metal software or a small real-time operating system. Their power consumption is relatively insensitive to computational load.	Pumpkin PPM family: 0.05 W, operating state not specified [2, p. 231] [3, p. 235] [4, p. 233]. AAC Clyde Space Kryten-M3: 0.4 W [2, p. 228] [3, p. 230] [4, Table 8-4]. C3S Electronics OBC: 0.46 W in test mode [2, p. 229] [3, p. 232] [4, p. 230]. SkyLabs NANOobc-2: < 0.9 W idle and < 1 W peak [2, p. 232] [3, p. 236].

Table 10: OBC power-consumption trends and reported operating states (Continued)

Platform Complexity	Reported Power Profile	Design Characteristics	Example Designs
Medium (Integrated OBC)	**Idle:** approximately 0.2–1 W **Nominal/steady:** approximately 1–3 W **Peak:** approximately 1.5–5 W	Typical integrated CubeSat OBCs combining command and data handling, mass-memory management, communication interfaces, fault protection, and basic payload management. Power may vary with processor utilization, interface activity, and memory access.	SPiN MA61C CubeSat: 1–1.2 W [2, p. 232] [3, p. 236] [4, Table 8-4]. AAC Clyde Space Sirius OBC & TCM: 1.3 W [2, p. 228] [3, p. 230] [4, Table 8-4]. EnduroSat OBC I: 0.2 W idle and 1.5 W peak [2, p. 229] [3, Table 8-1] [4, p. 230]. Resilient Computing RadPC: 1.5 W [2, p. 232] [3, p. 236] [4, p. 233]. SatRev OBC: 2.8 W typical and 5 W peak [4, p. 233].
High (Edge / AI / SoM)	**Idle/static:** approximately 5–15 W **Typical/active:** approximately 6–10 W **Peak/full load:** commonly 10–20 W, with larger payload computers reaching substantially higher values	Heterogeneous systems-on-module and payload-processing computers supporting image processing, software-defined radio, autonomous navigation, data compression, and machine-learning inference. Power is strongly dependent on processor utilization, FPGA configuration, accelerator activity, and payload duty cycle.	Novo Space GPU002AF: 6–11 W in active mode [2, p. 231] [3, p. 235] [4, p. 233]. KP Labs Leopard DPU: 7 W static and up to 10 W during deep-learning inference [2, p. 230] [3, p. 234] [4, p. 231]. Aitech S-A1760: $\leq$ 5 W idle, 8–10 W under a typical CUDA load, and $<$ 20 W at full utilization [2, p. 228] [3, Table 8-1] [4, Table 8-4]. EnduroSat GPC: $<$ 15 W idle and 130 W peak [4, p. 231].

Based on these results, a nominal power-design envelope of 1–10 W is adopted for the proposed OBC. This range encompasses conventional integrated CubeSat OBCs and moderate edge-processing workloads, while transient loads above 10 W are treated separately as peak operating conditions. The selected range is a project design boundary rather than the full commercial-hardware envelope, which extends from approximately 0.05 W for low-power microcontroller boards to 130 W peak for high-performance payload computers.

[TODO: peak loads and durations]

Protections

Protecting the electrical compenets of the BUS electronics from the harsh space environment is of paramount concenr to the design. This section details the consideraitons with respect to the design and how to achieve them

Table 11: Protection considerations

Consideration	Description	Example solutions
Mission radiation environment	Radiation exposure depends on orbit, inclination, mission duration, shielding, and solar activity. Parts should be selected against a mission-specific radiation analysis with an appropriate design margin [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Model the expected environment using tools such as SPENVIS [49] or CREME-MC [50] and select components whose test conditions exceed the predicted exposure. Athena uses a TMS570 device characterized to approximately 30 krad, while ORCASat selected a device tested to 6 krad [51, 8, 9].

Table 11: Protection considerations (Continued)

Consideration	Description	Example solutions
Total ionizing dose (TID)	TID represents cumulative ionizing-radiation exposure and can gradually alter device parameters or cause permanent failure. NASA notes that many COTS parts may tolerate only approximately 2–10 krad, although tolerance varies strongly with device and fabrication process [1, pp. 177–178] [2, pp. 191–193] [3, pp. 194–196] [4, pp. 191–192].	Use components with tested TID ratings, such as the VORAGO VA41630 MCU or GR712RC processor, or apply shielding where COTS components cannot meet the required dose [52, 53].
Displacement damage dose (DDD)	Displacement damage results from non-ionizing particle interactions and must be considered separately from TID, particularly for sensors, optoelectronics, and some semiconductor technologies [1, p. 219] [2, p. 235] [3, pp. 240–241] [4, p. 221].	Require component-specific proton or neutron characterization rather than assuming that a TID rating also covers displacement damage. Microchip publishes separate proton, TID, and SEE characterization for its radiation-tolerant FPGA families [54].
Single-event upset (SEU)	An SEU changes a memory or logic state without necessarily causing permanent damage. Protection is required for processor state, working memory, non-volatile memory, and FPGA configuration [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Use ECC or EDAC, memory scrubbing, redundant boot images, software consistency checks, and configuration-immune programmable logic. The GR712RC provides memory EDAC, while the RTG4 FPGA uses SEU-immune flash configuration [53, 55].
Single-event latch-up (SEL)	SEL may place a semiconductor device into a destructive high-current state. The affected circuit must be detected and disconnected before permanent damage occurs [1, pp. 219–220] [2, pp. 235–236] [3, pp. 240–241] [4, pp. 221–222].	Select SEL-immune components where practical and place vulnerable loads behind current-limited, resettable power switches. The 3D PLUS 3DPM0168 LCL provides overcurrent detection, load disconnection, and controlled repowering [52, 56].
COTS versus radiation-hardened components	COTS components generally provide higher performance, lower cost, and better availability, while radiation-hardened parts provide greater environmental assurance. Small spacecraft frequently use a hybrid approach [1, p. 219] [2, p. 235] [3, p. 240] [4, pp. 221–222].	Combine a COTS processor with radiation-tolerant supervisors, memories, programmable logic, or power-control circuitry. NEUDOSE uses a COTS AVR32 primary processor with a secondary Zynq-7000 system designed using both radiation-hardened and non-radiation-hardened components [10].
Watchdog and fault recovery	Radiation-induced faults may leave the processor or communication interface unresponsive. Recovery mechanisms should remain operational even when the primary processor has failed [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 219–220,222].	Use an independent watchdog, reset supervisor, communication timeout, and individually switchable power domains. The GR712RC includes watchdog support, while external protection switches can independently cycle affected loads [53, 56].
Memory protection	Processor memory, mass storage, boot images, and FPGA configuration can have different radiation responses and should be assessed independently. NASA identifies ECC, EDAC, scrubbing, and redundant storage as common mitigation methods [1, pp. 218–220] [2, pp. 234–236] [3, pp. 239–241] [4, pp. 221–224].	Use ECC-protected working memory, radiation-tolerant MRAM, redundant boot images, and TMR-protected non-volatile storage. Example devices include GR712RC EDAC-protected memory and radiation-tolerant MRAM and SPI NOR modules [53, 57, 58].

Table 11: Protection considerations (Continued)

Consideration	Description	Example solutions
Redundancy and voting	Redundancy may be applied at the logic, processor, memory, board, or subsystem level. It improves fault tolerance but increases power, mass, synchronization, and verification requirements [1, p. 220] [2, p. 236] [3, p. 241] [4, pp. 219–222].	Use lockstep processors, cold-redundant OBCs, triple-modular redundancy, or distributed backup controllers. Athena and ORCASat use lockstep-capable safety MCUs, while 3D PLUS offers TMR-protected SPI NOR memory [8, 9, 58].
Radiation-hardened-by-design (RHBD)	RHBD improves radiation tolerance through hardened circuit cells, radiation-aware layout, specialized device structures, and selective hardening of vulnerable functions [4, p. 222].	Use RHBD devices such as VORAGO HARDSIL-based Cortex-M MCUs or Microchip RTG4 FPGAs with radiation-hardened registers and flash-based configuration [52, 55].
Structural and localized shielding	The spacecraft structure provides some inherent shielding, but protection depends on material, thickness, component placement, and shielding in every direction. Local shielding may be more mass-efficient than increasing the thickness of the entire structure [1, pp. 176–179] [2, pp. 190–194] [3, pp. 193–196] [4, pp. 190–193].	Place sensitive processors or memories within aluminium, titanium, graded-Z, or composite shields. Commercial localized shielding products, such as Plasteel, target high-performance COTS processors, cameras, and memories [59, 60].
Secondary-radiation effects	Increasing shield thickness does not always proportionally reduce dose. High-energy particles interacting with shielding may generate secondary particles, making material choice and layer ordering important [1, pp. 176–179] [3, pp. 193–196] [4, pp. 190–193].	Evaluate shielding using particle-transport simulation rather than aluminium thickness alone. Graded-Z shielding combines materials with different atomic numbers to reduce secondary radiation [60].
Surface charging and electrostatic discharge	Surface charging and electrostatic discharge can damage or disrupt electrical systems. This is separate from TID and single-event-effect protection [1, p. 179] [2, p. 194] [3, pp. 196–197] [4, p. 193].	Use conductive bonding, controlled grounding paths, charge-dissipative coatings, and appropriate PCB surface treatments. The Shields-1 experiment included a charge-dissipation film together with radiation-shielding technologies [60].

Conclusion

This report outlined the state-of-the-art OBC review from both internal design stadpoint through the Design Space section as well as system integration standpoint through System Architecture section. Internal design is broken into selection of control unit, protocol set and interconnect topology, as well as periptheral array. Each of these compoenents have a wide selection of compoenetns and designs which will need to be researched further and used as part of trades once requirements are established. Further research and collaboration with other subsystems is also required to guide the internal design decisions for this subsystem.

Appendix A: Academic OBCs

Name	Team	MCU	Protocols	Standard	Notes	Ref
STM32L4 PC/104 academic OBC	VST104 thesis design	STM32L4—ARM-based	CAN	PC104	pc 104 cubesat design.	[5]
Athena OBC	University of Alberta (Albertasat)	TI TMS570LC4357 (ARM Cortex R5F)			Embedded memory type: Flash. Features ideal for satellite OBC application, including radiation & space reliability (rad-hard, SEL immune 48 MeV at 125°C, LAT 30krad), high computational performance (lockstep dual-core compare, 300MHz CPU clock, 1.66DMIPS/MHz).	[6, 8]
Orcasat OBC	University of Victoria	TI TMS570LS0714 (Dual ARM Cortex-R4F)			Embedded memory type: Flash. Operates in lockstep, ECC integrated flash & RAM banks, radiation-tested up to 6 krad, 80MHz core clock.	[7, 9]
LORIS OBC	Dalhousie University	NXP i.MX6DL (Single-core ARM Cortex-A9)	USB 2.0; SPI; I2C; PWM; UART; GPIO; ADC		Embedded memory type: Flash. Low profile, power efficient SoC SBC. 4GB eMMC flash storage, 256MB DDR3 RAM, 800MHz operational frequency, -40°C to 85°C temperature range.	[11, 34]
NEUDOSE OBC	McMaster University	Primary: AVR32. Secondary: Xilinx Zynq-7000 SoC			Primary is COTS chosen for high TRL to reduce mission risk; secondary is designed for radiation tolerance using rad-hard and non-rad-hard components.	[10]
RADSAT-SK OBC	University of Saskatchewan	ARM926EJ-S			[29]	

Appendix B: Commercial OBC Payload Integration Survey

Table 14 surveys commercial OBC and payload computing platforms with respect to data and control communication throughputs, volatile/non-volatile buffering, transfer mechanisms, and event notification interfaces. The performance and specification confidence level of figures listed in Table 14 are categorized according to the notation in Table 13.

Table 13: Throughput and Performance Data Notation

Mark	Meaning
[M]	Measured end-to-end or application-level throughput
[S]	Product-specific rate stated by the manufacturer
[N]	Nominal line rate implied by an interface name, such as Gigabit Ethernet
[L]	Latency figure rather than throughput
[U]	Undisclosed — interface is present, but its usable or configured rate is not published

Table 14: Commercial OBC and Payload Computer Integration Survey

OBC or Payload Computer	Payload Data Throughput	Control Throughput & Event Interfaces	Volatile & Non-Volatile Buffering	Transfer, Request, & Notification Mechanism
GomSpace NanoMind HP MK3 [61]	Three independent full-duplex SpaceWire payload links. Vendor iperf3 tests at the network layer with MTU 9000 report 150 Mbit/s for one unidirectional interface [M], 150 Mbit/s aggregate for three unidirectional interfaces [M], and 80 Mbit/s aggregate for three bidirectional interfaces [M]. RS-422 TCP throughput is 3 Mbit/s through the processing system [M] and 2.2 Mbit/s through programmable logic [M]. Storage tests report approximately 23 MB/s read and 21 MB/s write for one eMMC [M], or 45 MB/s read and 41 MB/s write when striped [M].	Two CAN buses, I^2C, differential PPS and reference-clock inputs, RS-422, and USB 2.0 host/device are exposed, but public CAN and I^2C rates are not stated [U]. The supplied I^2C software supports attached slave devices but not multi-master operation or CSP over that interface.	1 GB DDR3, reduced to 512 MB usable data memory when ECC is enabled; 256 MB NOR flash; 70 GB pSLC eMMC and 233 GB MLC eMMC. The storage can be configured using logical-volume arrangements such as striping or mirroring. These are installed capacities, not stated payload reservations.	SpaceWire endpoints are implemented in the Zynq programmable logic. Payload data is presented through network interfaces and software drivers, and the datasheet separately benchmarks processing-system and programmable-logic RS-422 paths. The Zynq architecture permits memory transfers between programmable logic and system memory, but the product datasheet does not disclose the shipped DMA-channel mapping, descriptor format, queue ownership, cache-coherency policy, or whether the SpaceWire software path is zero-copy.

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
	SpaceWire: 80–150 Mbit/s [M] **RS-422: 2.2–3 Mbit/s [M]** **Storage: 21–45 MB/s [M]**		**Volatile: 512 MB–1 GB [S]** **Flash: 256 MB [S]** **Mass Storage: 70–233 GB [S]**	
AAC Clyde Space Sirius QuadCore [62]	Four SpaceWire ports with a vendor figure of "up to 300 Mbit" through the inbuilt router [S]. The public document does not say whether this is per-port throughput, aggregate router throughput, or a nominal configuration limit.	Two CAN 2.0B buses are used for low-level commanding and monitoring [U]. The unit also exposes an RS-422 UART, SPI, and nine GPIO signals, without published rates. AAC explicitly describes CAN as the low-level control path and the SpaceWire router as the fast payload/subsystem path.	4 Gbit SDRAM for data plus 2 Gbit for EDAC, 2 Mbit MRAM boot memory, 256 Mbit SPI flash image memory, and a 2 MiB L2 cache. The product sheet does not identify a payload-reserved fraction.	This is a clear dual-plane architecture: payload data travels through the SpaceWire router, while CAN carries commands and monitoring messages. Software may run Linux or RTEMS. No public information identifies whether SpaceWire-to-SDRAM movement uses a dedicated DMA engine, CPU copies, router-local buffering, or a mixture of these.
	SpaceWire: Up to 300 Mbit/s [S]		**Volatile: 512 MB [S]** **Flash: 256 Kbit–32 MB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	Payload Data Throughput	Control Throughput & Event Interfaces	Volatile & Non-Volatile Buffering	Transfer, Request, & Notification Mechanism
AAC Clyde Space Sirius TCM LEON3FT [63, 64, 26]	Two SpaceWire interfaces at 50 Mbit/s [S]. The development-kit datasheet explicitly lists RMAP support [26], making this one of the clearer commercial examples of remote payload-memory access over SpaceWire.	Three RS-422/485 UARTs, two additional RS-485-only UARTs, an RS-485 PPS input, twelve GPIO, S- and X-band transceiver command/housekeeping channels, and twelve RS-422 pulse-command outputs are provided; serial baud rates are not disclosed [U].	64 MB SDRAM after EDAC, 16 kB NVRAM after EDAC, 2 GB system NAND after EDAC, and 32 GB mass-memory storage after EDAC.	The TCM is specifically described as receiving and storing payload and housekeeping data, distributing telecommands, and serving mass-memory data to the transceiver. With RMAP [26], another SpaceWire node can issue reads and writes to exposed memory, registers, FIFOs, or mailboxes; pulse outputs provide an independent low-level command/event mechanism. The public material does not expose the actual RMAP address map, access-control rules, maximum transaction size, queue depth, or number of simultaneously outstanding transactions.
	SpaceWire: 50 Mbit/s [S]		**Volatile: 64 MB [S]** **Flash: 16 kB–2 GB [S]** **Mass Storage: 32 GB [S]**	
EnduroSat Onboard Computer [65, 24]	100BASE-TX Ethernet → 100 Mbit/s nominal [N]. CAN-FD, RS-422, RS-485, SPI, I^2C, and UART payload interfaces are present, but their configured or achieved product-level rates are not published [U].	Two CAN buses operate at 1 Mbit/s using CSP-ES [S] [24]. Sixteen payload enable, disable, and feedback GPIO signals and a single-ended PPS signal provide discrete control, status, and synchronization paths.	2 MB program flash, two 16 GB embedded SD-NAND devices, and two 2 MB FRAM devices with ECC capability. The public page does not state the volatile SRAM capacity or how much NAND/FRAM is available to payload applications.	Packetized command and control is supplied through CSP-ES over CAN [24]. The flight-management software includes drivers, housekeeping, logging, scheduling, tasking, compression, and archiving, while direct GPIO provides payload power-control and feedback events. No public DMA topology, queue API, payload file-service API, or interrupt-completion model is documented.
	Ethernet: 100 Mbit/s [N]	**CAN (CSP): 1 Mbit/s [S]**	**Flash: 2 MB [S]** **Mass Storage: 32 GB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
Spacemanic Deep Thought [66, 67, 24]	External SPI is typically 40 MHz [S]. RMII and USB signals are exposed, but the product datasheet gives no achieved Ethernet or USB throughput [U].	CSP over CAN up to 1 Mbit/s [S] [24]; CSP over I^2C up to 400 kHz [S]; CSP over RS-485/KISS up to 115,200 baud [S]; isolated I^2C up to 400 kHz [S]. GPIO, PWM, ADC, clock, and PPS-capable pins provide discrete events and timing.	The SAMV71 family [67] provides up to 384 kB on-chip SRAM. The board includes three 1 MiB FRAM devices and four 32 MiB flash devices; Spacemanic explicitly says all seven external memories are available for mission-specific software.	CAN is implemented using the SAMV71's integrated MCAN peripheral followed by an external CAN transceiver; it is not a separate external CAN-controller IC. SPI, I^2C, GPIO, and RMII signals connect directly to the MCU. The SAMV71 [67] silicon includes a 24-channel central XDMAC and MCAN controllers with SRAM-based mailboxes, so DMA-assisted peripheral transfers are architecturally possible; however, the OBC datasheet does not say which supplied drivers use DMA or whether CAN frames are copied from message RAM by the CPU.
	SPI: 40 MHz [S]	**CAN (CSP): 1 Mbit/s [S]** **I^2C (CSP): 400 kHz [S]** **RS-485/KISS: 115.2 kbaud [S]**	**Volatile: Up to 384 kB [S]** **Flash: 3 MB–128 MB [S]**	
Spacemanic Eddie [68, 24]	External payload SPI up to 16 MHz [S].	CSP over CAN up to 1 Mbit/s [S] [24]; CSP over I^2C up to 400 kHz [S]; CSP over RS-485/KISS up to 38,400 baud [S]. GPIO, PWM, ADC, PPS, and a pin labelled DMA TRIG are exposed.	No separate large volatile payload RAM is advertised. Six 1 MB FRAM devices are all available to mission software, and the example firmware can combine them as one FatFs disk.	Unlike Deep Thought, Eddie uses an external MCP25625 CAN controller connected to the MCU through SPI. The controller provides an interrupt signal and receive-buffer-full signals, which notify the MCU that CAN data or an event requires servicing. The public datasheet identifies a DMA TRIG connector pin but does not define its transaction protocol, destination buffer, handshake, or relationship to the CAN controller.
	SPI: 16 MHz [S]	**CAN (CSP): 1 Mbit/s [S]** **I^2C (CSP): 400 kHz [S]**	**Flash: 6 MB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
		RS-485/KISS: 38.4 kbaud [S]		
Xiphos Q8/Q8S [69]	Gigabit Ethernet gives 1 Gbit/s nominal [N]. USB 3.1 Gen 1, USB 2.0, 64 LVDS pairs, and three gigabit transceivers for interfaces such as SATA or PCIe are available, but achieved payload throughput is not published [U].	A CAN controller is implemented in programmable logic. Factory-selectable RS-232/422/485, GPIO/MIO, and RTC wake/alarm interrupt facilities are also provided, without public rates [U].	4 GB LPDDR4 with EDAC, two 256 MB QSPI NOR devices, and two independent 128 GB eMMC devices with separately controlled buses and power.	Payload interfaces are accessed through standard Linux/Xilinx drivers or Xiphos custom drivers; programmable logic can host mission-specific interface and pre-processing logic. The public sheet does not disclose DMA topology, descriptor rings, payload buffer allocation, measured storage bandwidth, or whether the independent eMMC devices can be sustained concurrently at a stated rate.
	Ethernet: 1 Gbit/s [N]		**Volatile: 4 GB [S]** **Flash: 512 MB [S]** **Mass Storage: 256 GB [S]**	
Unibap iX10-101 [70]	Two 10GBASE-T ports → 10 Gbit/s per port nominal [N]. The unit also provides two Camera Link Base interfaces, two SpaceWire ports, two USB 3.0 ports, and one USB 2.0 port, with no achieved product-level rates published for those interfaces [U].	CAN 2.0/FD, two I^2C interfaces, PPS/proprietary PLET timing, one RS-232 port, and four RS-485/422 ports are provided without published configured rates [U].	24 GB DDR4 ECC associated with the CPU/GPU system, two 4 TB NVMe SSDs, and one 128 GB SATA SSD. These are installed capacities rather than stated payload reservations.	The vendor's architecture depicts sensors feeding FPGA-based data acquisition, followed by CPU orchestration, GPU/VPU processing, and storage used as both a buffer and long-term store before data is delivered to the spacecraft bus, OBC, or communications system. This is strongly suggestive of a DMA-oriented streaming architecture, but the public document does not identify the DMA engines, descriptor format, backpressure mechanism, or payload software API.
	Ethernet: 10 Gbit/s [N]		**Volatile: 24 GB [S]** **Mass Storage: 128 GB–8.1 TB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	Payload Data Throughput	Control Throughput & Event Interfaces	Volatile & Non-Volatile Buffering	Transfer, Request, & Notification Mechanism
KP Labs Antelope [71]	1 Gb Ethernet → 1 Gbit/s nominal [N]. LVDS or GTH transceivers, USB 3.0, USB 2.0, and SATA are also available, but achieved rates are not disclosed [U].	CAN, I^2C, SPI, RS-422/485, UART, GPIO, and GPS PPS are present without product-level rates [U].	8 GB DDR4 with ECC, 4 GB SLC NAND, and optional SATA SSD storage.	The Zynq UltraScale+ combines application processors, real-time processors, and FPGA fabric for custom functions; software can run 64-bit Linux or bare metal. This enables several possible payload-access models—device drivers, memory-mapped FPGA registers, DMA streams, or shared-memory queues—but the vendor does not publicly specify which are supplied as standard payload services.
	Ethernet: 1 Gbit/s [N]		**Volatile: 8 GB [S]** **Flash: 4 GB [S]**	
CesiumAstro SBC-1461 [72]	Two 10GBASE-KR ports → 10 Gbit/s each nominal [N], two 100BASE-TX ports → 100 Mbit/s each nominal [N], plus SGMII and USB 3.0 [U].	Two RS-422/485 channels, SPI with two chip-selects, UART, JTAG, and IEEE-1588 time synchronization are provided without public application-level rates [U].	2 GB ECC SDRAM, with an option for 4 GB or more; 8 GB eMMC; and redundant 64 MB boot flash with automatic failover.	The product advertises traffic routing and encryption without loading the processor cores, indicating hardware offload for portions of the data path, and includes a Linux SDK. However, the available detailed public sheet is dated 2022 and does not expose the payload API, DMA engines, queue organization, measured end-to-end throughput, or payload-reserved memory.
	Ethernet: 10 Gbit/s / 100 Mbit/s [N]		**Volatile: 2–4 GB [S]** **Flash: 64 MB [S]** **Mass Storage: 8 GB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
AAC Clyde Space Kryten-M3 [73]	Twenty LVDS lines, three QSPI interfaces—two using LVDS signalling and one using 3.3 V logic—and one SPI controller with seven chip-select lines. Product-level rates: [U]. The LVDS lines are physical signalling resources rather than a defined packet protocol in the public specification.	One CAN interface, two I^2C interfaces, eight UARTs, one RS-422 interface that can instead be configured as two RS-485 links, GPIO and DTMF. Rates: [U].	Volatile: embedded SRAM/cache capacity is not stated publicly [U]. Non-volatile: 16 MB MRAM and 4 GB SLC flash; the boot configuration is listed as 256 kB eNVM plus 8 MB MRAM.	SmartFusion2 integrates the Cortex-M3 and FPGA fabric. RTEMS, EDAC-protected memories, hardware recovery, Reed–Solomon and cryptographic acceleration are provided. Payload communication is therefore software/driver mediated, with FPGA logic available for custom interfacing. Public material does not disclose DMA request mappings, FIFO depths, interrupt semantics, or whether LVDS/QSPI transfers can be performed zero-copy.
			Flash: 16 MB [S] **Mass Storage: 4 GB [S]**	
Pumpkin PPM family [74]	No uniform family-wide data interface exists. Individual PPMs expose the selected MCU or DSP peripherals through the Pumpkin motherboard, including SPI, serial interfaces, GPIO and variant-specific external buses. Rates: [U]. Figures such as the D1's 32 MHz clock or 16 MIPS performance are processor figures, not payload-link throughput.	I^2C, UART, SPI, GPIO and hardware radio-handshake signals are available in various configurations. Availability and pin mapping depend on the PPM and motherboard combination. Rates: [U].	Volatile: MCU-internal RAM and motherboard-dependent expansion; no common family value. Non-volatile: MCU flash plus variant-specific storage. The PPM D1, for example, includes 64 Mbit (8 MB) of external serial flash; the flight motherboard can provide additional mass-storage functions.	The PPM is a replaceable processor module that maps MCU I/O to a common motherboard architecture. Payload access is consequently through ordinary MCU peripheral drivers, polling, interrupts or any DMA supported by the selected MCU. There is no family-wide published DMA, message-buffer or shared-memory architecture.
			Flash: Up to 8 MB [S]	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
C3S Electronics OBC [75, 76]	Four M-LVDS channels arranged as two cold-redundant pairs. C3S platform documentation identifies one redundant pair for communications equipment and one pair for the payload. M-LVDS rate: [U].	Two CAN interfaces forming one cold-redundant command-bus pair. CAN rate: [U].	Volatile: 384 kB internal RAM. Program storage: 2 MB internal flash. Non-volatile daughterboard: 16 MB radiation-resistant MRAM and 16 GB eMMC.	This is one of the clearest public examples of an explicit command/data split: CAN is used as the command bus and M-LVDS as the payload data bus. The interfaces and memories are cold-redundant at platform level. Public documentation does not state the M-LVDS framing protocol, flow control, DMA path, FIFO capacity, payload file API, or failover timing.
			Volatile: 384 kB [S] **Flash: 2–16 MB [S]** **Mass Storage: 16 GB [S]**	
SkyLabs NANOobc-2 [77]	Two "high-speed" LVDS interfaces, alternatively configurable as RS-422/RS-485. Verified product-specific rate: [U]. Some mirrored documentation reports 20 Mbit/s for LVDS [S], but that figure is not clearly stated on the current public manufacturer page.	Redundant CAN-TS for telemetry and telecommand; eight multifunction GPIOs can support UART, SPI, TWI and an onboard-time trigger. Rates: [U].	Published revisions/listings conflict [U]. Values include 1–2 MB SRAM, 2–4 MB MRAM, a 4 Mbit telemetry NVM, and redundant NAND described variously as 2 Gbit or 2 GB.	The NANOsky CMM software provides CAN-TS and LVDS-TS stacks, permanent-storage management, redundancy management, housekeeping acquisition, FDIR, software scrubbing and firmware update. The PicoSkyFT architecture includes prioritized interrupts and fault traps. Public sources do not specify DMA; the documented model is primarily interrupt-, driver- and FDIR-managed communication.
	LVDS: 20 Mbit/s [S]		**Volatile: 1–2 MB [S]** **Flash: 500 kB–4 MB [S]** **Mass Storage: 256 MB–2 GB [S]**	

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	Payload Data Throughput	Control Throughput & Event Interfaces	Volatile & Non-Volatile Buffering	Transfer, Request, & Notification Mechanism
SPiN MA61C [78, 79]	CubeSat and SmallSat configurations provide two SpaceWire interfaces. SpaceWire rate: [U]. The product is intended to route and translate payload and subsystem data rather than act only as a conventional processor board.	Depending on the configuration: CAN, I^2C, SPI, RS-422, RS-485, RS-232, GPIO and MIL-STD-1553B. The ESA presentation reports a system response time below 1 ms [L], but does not report sustained link throughput [U].	Current vendor page lists 64 Mbit (8 MB) SRAM and 64 Mbit (8 MB) flash. Some distributor material additionally lists 3 Gbit SDRAM.	MA61C stores device drivers and electronic data sheets, scans interfaces, recognizes connected equipment, self-configures, installs/selects drivers, buffers data, and routes or converts traffic between interfaces. Its software uses data-link, network and transport abstraction layers.
		Response Latency: <1 ms [L]	**Volatile: 8–375 MB [S]** **Flash: 8 MB [S]**	
Argotec FERMI OBC&DH [80]	SpaceWire and LVDS are publicly listed as payload-capable high-speed interfaces. Rates: [U].	RS-422, SPI, I^2C and GPIO are listed for lower-rate equipment and control integration. Rates: [U].	A current secondary product listing gives 256 MB SDRAM, 20 Mbit plus 5 Mbit EEPROM, and 16 GB ECC-corrected mass memory.	A dual-core LEON3FT, fault-tolerant memory controller and Microchip RTG4 FPGA combine software processing with FPGA acceleration. Public sources do not identify DMA channels, SpaceWire receive-buffer depths, descriptor formats, interrupt routing or payload-access APIs.
			Volatile: 256 MB [S] **Flash: 3.1 MB [S]** **Mass Storage: 16 GB [S]**	
SatRev OBC [81, 2]	Interfaces and rates: [U]. SatRev's public website currently emphasizes complete spacecraft and hosted payload services rather than exposing an OBC product datasheet.	Interfaces and rates: [U].	Volatile and non-volatile memory: [U].	NASA's survey identifies a quad-core Cortex-A72 processor at 1.5 GHz and gives board-level power figures, but does not provide a payload ICD, interface list, buffer architecture or transfer mechanism.

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	**Payload Data Throughput**	**Control Throughput & Event Interfaces**	**Volatile & Non-Volatile Buffering**	**Transfer, Request, & Notification Mechanism**
Resilient Computing RadPC-SBC-001 [4]	Payload interfaces and rates: [U].	Control interfaces and rates: [U].	Volatile and non-volatile memory: [U].	NASA describes a 32-bit RISC-V processor implemented in an AMD Artix-7 FPGA with COTS single-event-effect mitigation. No public payload-interface map, DMA implementation, shared-memory model or event-notification mechanism was located.
Andes OBC [3, p. 236][4, p. 250]	Public commercial listings mention I^2C, SPI and UART, but no clearly identified high-rate payload link or numeric rate was located. Rates: [U].	I^2C, SPI and UART can provide command/control integration. Rates: [U].	Full cold redundancy is stated for the OBC, MRAM and NAND, but installed memory capacities are not disclosed publicly [U].	Two Cortex-M7 processors and duplicated MRAM/NAND implement cold redundancy. Public information does not say whether each processor has a separate interface controller or how buses are switched during failover.
Aerospacelab OBC [3]	The standalone OBC's public NASA entry does not state payload interfaces or rates. A later Aerospacelab presentation lists configurations with up to 40 digital interfaces—including Ethernet and SpaceWire—rates: [U].	The centralized-avionics concept lists CAN, RS-422 and RS-485 among its control interfaces. Rates: [U].	Memory capacities: [U].	The documented processor is a Zynq-7045 containing dual Cortex-A9 cores and Kintex-7 FPGA fabric, supervised by a rad-hard MCU and hardware watchdog. No public DMA, buffering, payload-request or notification specification is available.
Bradford Binary OBC [2]	Payload interfaces and rates: [U]. No current Binary product page or public datasheet was located in Bradford's current catalogue.	Control interfaces and rates: [U].	Memory capacities: [U].	The NASA 2023 table describes dual Cortex-R5 processors, software EDAC and internal cold redundancy, with approximately 3 W peak and 1 W idle power. The current Bradford portfolio promotes the Stellar system, suggesting Binary is legacy.

Table 14: Commercial OBC and Payload Computer Integration Survey (Continued)

OBC or Payload Computer	Payload Data Throughput	Control Throughput & Event Interfaces	Volatile & Non-Volatile Buffering	Transfer, Request, & Notification Mechanism
Bradford Stellar OBC and Payload Data-Handling System [82, 4]	Bradford describes Stellar as a scalable flight and payload data computer, but the public page does not publish interface types, port counts or rates. Throughputs: [U].	Control interfaces and rates: [U].	Volatile and non-volatile capacities: [U].	NASA identifies dual PolarFire SoCs, software EDAC and internal cold redundancy. The product combines spacecraft control and payload data handling, but omits the memory map, failover, DMA paths and payload software interface.

Appendix C: PC/104 Pinout Survey

Table 15 surveys PC/104 header pinout assignments across ISO standards, open specifications, and commercial OBCs/peripherals.

Table 15: PC/104 Pinout Survey Across Standards and Commercial Platforms

Specification / Platform	PC/104 Pinout	Source
ISO 17981:2024 — CubeSat interface	H1-41 — SDA H1-43 — SCL H2-25/26 — 5V H2-27/28 — 3.3V H2-29...32 — GND	[83, 84]
LibreCube Board Specification	H1-1 — CAN_A_L H1-2 — CAN_B_L H1-3 — CAN_A_H H1-4 — CAN_B_H H1-5/7 — UART_A H1-6/8 — UART_B H1-13/15 — UART_C H1-22/24 — UART_D H1-25/26 — 5V_PERM H1-31/32 — CHARGE H1-33/34 — GND H1-38/40 — UART_E payload H1-42/44 — UART_F payload H1-45...52 — PWR_ENA H2-1...24/33...40 — PWR_ENA H2-25/26 — 5V H2-29...32 — GND H2-45/46 — battery H2-49 — RBF_A H2-50 — RBF_B H2-51 — deployment switch A H2-52 — deployment switch B	[85]

Table 15: PC/104 Pinout Survey Across Standards and Commercial Platforms (Continued)

Specification / Platform	PC/104 Pinout	Source
Pumpkin CubeSat Kit Bus / MBM2	H1-1…24 — IO H1-25 — -FAULT H1-26 — VREF0 H1-27 — current SENSE H1-28 — VREF1 H1-29 — -RESET H1-30 — VREF2 H1-31 — OFF_VCC H1-32 — +5V_USB H1-33 — PWR_MHX H1-34 — -RST_MHX H1-35 — -CTS_MHX H1-36 — -RTS_MHX H1-37 — -DSR_MHX H1-38 — -DTR_MHX H1-39 — TXD_MHX H1-40 — RXD_MHX H1-41 — SDA_SYS H1-42 — VBACKUP H1-43 — SCL_SYS H1-44/45 — reserved H1-46 — PPS H1-47…52 — USER H2-1…24 — IO H2-25/26 — +5V_SYS H2-27/28 — VCC_SYS H2-29/30/32 — DGND H2-31 — AGND H2-33…44 — S H2-45/46 — VBATT H2-47…52 — USER	[86]

Table 15: PC/104 Pinout Survey Across Standards and Commercial Platforms (Continued)

Specification / Platform	PC/104 Pinout	Source
EnduroSat OBC Type I/II — legacy model	H1-1 — CANL H1-3 — CANH H1-19 — PAY_TX H1-20 — PAY_RX H1-21 — PAY_SCL H1-23 — PAY_SDA H1-33 — UHF RX H1-35 — UHF TX H1-39 — SYS_TX H1-40 — SYS_RX H1-41 — SYS_SDA H1-43 — SYS_SCL H2-3 — OBC_OUT1 H2-4 — OBC_OUT2 H2-5 — OBC_OUT3 H2-6 — EN_OBC H2-7 — OBC_OUT5 H2-8 — OBC_OUT6 H2-9 — SPI MISO H2-10 — SPI MOSI H2-11 — SPI SCK H2-13 — SPI TR H2-15 — SPI CS H2-16 — SPI IRQ H2-25/26 — +5V H2-27/28 — +3.3V H2-29...32 — GND H2-47...50 — payload SPI	[87, 88]

Table 15: PC/104 Pinout Survey Across Standards and Commercial Platforms (Continued)

Specification / Platform	PC/104 Pinout	Source
Simera Sense TriScape100 Control Electronics	H1-1...3/17...20/22/24/30/31/33/40 — IO H1-4 — IO / PowerCtrl / LVDS6- H1-5/6 — IO / PowerCtrl / LVDS6+ H1-7/8 — IO / PowerCtrl / LVDS7+ H1-9/10 — IO / PowerCtrl / LVDS7- H1-11 — IO / PowerCtrl / LVDS3- H1-13/14 — IO / PowerCtrl / LVDS3+ H1-15/16 — IO / PowerCtrl H1-21/43 — IO / SCL H1-23/41 — IO / SDA H1-29 — IO / LVDS5- H1-32 — IO / LVDS5+ H1-35 — IO / LVDS2- H1-39 — IO / LVDS2+ H1-45 — 1-Wire VCC H1-46 — 1-Wire data H1-47/49/51 — 5 V opt H2-13/15/16/25/26 — 5 V opt H2-17/18 — IO / PowerCtrl / LVDS4- H2-19/20 — IO / PowerCtrl / LVDS4+ H2-21/22/48/49 — IO H2-29/30/32 — +5 V return H2-47 — IO / LVDS1- H2-50 — IO / LVDS1+	[89]

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